

WEEKLY PROGRESS REPORT – WEEK 4

Project Title

Healthcare Analytics Pipeline for MediCare Hospital

Group Number: 3

Team Members

- Arpit Sharma (G25AI1012)
 - Ankita Dwivedi (G25AI1011)
 - Astosh Ranjan (G25AI1013)
 - Bharat Sharma (G25AI1014)
 - Damini Khatri (G25AI1015)
-

Week 4 Objectives

The objective of Week 4 was to transform healthcare data for analytics, implement big data processing techniques, design the healthcare data warehouse, develop analytical queries, and optimize query performance to support business intelligence and reporting requirements.

Activities Completed

1. PySpark Data Processing

The healthcare dataset was processed using Apache PySpark to perform scalable data transformation and analytics operations. Various transformation techniques such as filtering, aggregation, and data cleaning were implemented to prepare the dataset for analytical processing.

2. Databricks Validation

PySpark processing workflows were validated using Databricks notebooks. The transformed datasets and analytical outputs were verified to ensure consistency and demonstrate industry-standard big data processing practices.

3. Data Warehouse Design

A healthcare data warehouse was designed using the Star Schema approach to support analytical reporting and decision-making requirements. The warehouse structure was planned to provide efficient storage and querying of healthcare data.

4. Fact and Dimension Table Development

Fact and dimension tables were created to organize healthcare information into analytical structures.

Dimension Tables:

- dim_patient
- dim_disease

- dim_date

Fact Table:

- fact_patient_visits

These tables were designed to support efficient reporting and analytical operations.

5. SQL Analytics Development

Various SQL analytical queries were implemented to generate business insights from the healthcare data. The queries were developed to analyze patient information, treatment costs, disease patterns, and hospital statistics.

6. Query Optimization

Database optimization techniques were applied to improve analytical query performance. Proper indexing and query optimization methods were considered to enhance execution efficiency and reporting performance.

7. Team Coordination and Review

The team conducted review sessions to validate transformed datasets, verify analytical outputs, discuss implementation progress, and prepare for the machine learning and dashboard development phases.

Deliverables Completed

- PySpark data processing pipeline
- Databricks notebook validation
- Healthcare data warehouse design
- Fact and dimension table implementation
- SQL analytical queries
- Query optimization activities
- Data warehouse documentation

Challenges Faced

- Managing large-scale healthcare data transformations
 - Designing an efficient Star Schema data warehouse
 - Optimizing analytical SQL queries
 - Validating transformed data and analytical outputs
 - Ensuring consistency across different processing environments
-

Plan for Week 5

- Perform machine learning analysis on healthcare data.
 - Develop an interactive Streamlit dashboard.
 - Implement Kafka demonstration for streaming concepts.
 - Configure CI/CD automation and deployment activities.
 - Conduct final testing, validation, and project documentation.
-

Conclusion

Week 4 primarily focused on healthcare data transformation, big data processing, and analytical data warehousing. The team successfully implemented PySpark-based data processing workflows, validated transformations using Databricks, designed the healthcare data warehouse, and developed analytical SQL queries. These activities established a scalable analytical foundation for the machine learning and visualization phases of the Healthcare Analytics Pipeline project.
